

Trading, storage, or penalty? Uncovering firms' decision-making behavior in the Shanghai emissions trading scheme: Insights from agent-based modeling

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ABSTRACT

The emissions trading scheme (ETS) has become a flagship climatic initiative for regulating greenhouse gas (GHG) emissions. Under an ETS, the emitting firm must simultaneously deal with changing carbon prices and the number of permits and the trade-off between permit trading (if one should buy, sell, or reserve) and permit-consuming production. This study characterizes the different compliance strategies' trade-offs, logic, and options among regulated emitting firms under an ETS. Consequently, this study provides firm-level evidence of regulated firms' strategic responses under the Shanghai ETS, considering detailed firm information, heterogeneous industry characteristics, and ETS architects. The proposed model simulated the trading interactions among diversified firms to portray the firms' decision-making process, and the emergent effects on the ETS market were identified. The results indicate that: 1) the carbon price experiences a non-monotonic "L-shaped" trend, which maintains an initial low level and increases sharply after crossing a threshold; 2) with increasing carbon prices, the trading in ETS becomes more active, especially among the low-emission firms; 3) the current ETS penalty is too limited and generic, which inadequately induces technological development and carbon reduction among firms. Finally, policy suggestions are provided for future optimization of the ETS mechanism. Overall, this study contributes to operation management literature by evaluating decision-making behaviors in a dynamic environment. Our findings have global implications for policymakers and managers in the private sector.

1. Introduction

As one of the most prominent policy instruments for promoting greenhouse gas (GHG) mitigation, the emission trading scheme (ETS) has received widespread popularity and growing desirability compared to traditional command-based alternatives (Zhu et al., 2019; Tietenberg, 2010; Huang et al., 2022). Drawing on ex-post evidence of the various comparative advantages of ETS, such as flexibility and comprehensiveness, a predictable and transparent market environment, and cost-effectiveness (Newell and Siikamäki, 2014; Tang et al., 2017; Zhu et al., 2019; Wei et al., 2021), there is a growing consensus that ETS is

the "most effective climate policy" to mitigate global GHG emissions (Tang et al., 2020; Meng et al., 2022). Recently, renewed interest in ETSs has spiked because of the expectation that ETS carbon prices will continue to increase as governments adopt more ambitious targets for climate conservation.

The introduction of ETSs encourages profit-seeking firms to regulate emissions by incentivizing efficiency increase and emission curbing. The ETS initiative aims to internalize the external cost of GHG emissions by sending price signals, thus incentivizing a shift to low-carbon actions (Cui et al., 2014). The ETS is generally composed of four leading players: the government (ETS designer), regulated firms (ETS target), third-party

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organizations (ETS regulators), and trading platforms (ETS platform) (Tang et al., 2020). The government initially sets the emission allowance over a specified compliance period. The emission quotas are the quantities of emissions that firms are allowed to emit, which are then allocated to specific regulated emitting firms (entities) in the ETS. Therefore, emission permits, such as capital, materials, and other commodity inputs, constitute a scarce factor in a firm's production function. Consequently, firms trade with their peers in the carbon market based on emission quota holding and abatement costs. At the end of the compliance period, the third-party organization audits the emission amount of each firm. If emissions exceed the firm's quota, a high fine is paid by the firm.

Notwithstanding the laudable policy objectives of ETS, which are to enable emission reduction, the potential of ETS is still largely untapped (The World Bank, 2021). For example, although ETS can help address price barriers for fighting climate change, the majority of ETS carbon prices are well below the basic levels (i.e., USD 40–80/tCO₂) to promote significant decarbonization and to meet the 2 °C goals of the Paris Agreement (e.g., Bayer and Aklin, 2020). Transaction volumes in most early ETSs were low and inactive, resulting in poor market liquidity. More broadly, emission reduction is far short of what is needed to achieve the Paris Agreement's temperature objectives (The World Bank, 2021). The complexity of ETS leading to less-than-expected market responses and performances has been addressed in the literature as one of the most challenging themes in energy and climate economics research (Tang et al., 2020; Sandoff and Schaad, 2009). The complexity of an ETS lies in three aspects: a variety of uncertain factors (Tang et al., 2015; Zhu et al., 2018), a series of rules and designs (Tang et al., 2017), and heterogeneous participating agents and their interactions (Tang et al., 2015; Wei et al., 2022a). The factors mentioned above jointly determine ETS policies' final efficacy and effects. Because firms dealing with emissions permits are free to trade in the market while fulfilling their production obligations, the core of the relevant discussions is that it is not well understood how firms make dynamic decisions in ETS.

At the micro level, firms wrestle with how to control the emission-regulated environment. Such decisions determine how firms trade off the investment costs of emission reduction, benefits from the commodity market, and the timing, quantity, and uncertain prices of permit trading. To achieve regulatory accomplishments, firms in the ETS have three options: independent emission reduction by low-carbon technologies, output adjustment, or quota trading (Fan, 2018). A firm's trading strategies consist of three options: buying, postponing/reserving, and selling allowances (Wei et al., 2022b). These decisions jointly affect the demand and supply structures of the carbon allowance market (Yu et al., 2020). Drawing on the architects of an ETS policy, a firm must balance various factors, including commodity production, allowance holding, marginal abatement costs for low-carbon technology, and expectation of future carbon prices. These factors make a firm's decision a complex trade-off. For example, which is more beneficial in ETS, buying or selling? What will be the price trend, and should the firm sell extra carbon permits in the current period or in the future?

At the macro level, the emergence properties from such disaggregated firm strategies lead to dynamic interaction and equilibrium among the ETS, commodity markets, and low-carbon policy. Particularly, amidst these options, firms' behaviors have a direct influence on ETS performance and may cause market failures, shown as significant volatility (Lyu et al., 2020), inactive market liquidity (Chang et al., 2018), timing-concentrated transition (Zhu et al., 2018), and sustained low prices (Ervin, 2018), which were experienced in a majority of early ETSs. Overall, the complexity of decision-making leads to substantial challenges for firms and brings high uncertainty to ETSs.

The research questions are as follows: What are firms' decision-making behaviors (i.e., purchasing, buying, storing, or suffering penalties) in an ETS with heterogeneous characteristics? What are the equilibrium performances of the ETS, including permit prices, trading volumes, and firm decisions? What are the government's strategies for

achieving the carbon reduction targets in the ETS? Previous literature mainly adopts equilibrium-based models, which make a series of simplifying assumptions, including the representative agent, market equilibrium status, rational expectations, and perfect complete (Yu et al., 2020). These assumptions limit the model, preventing it from fully considering the firms' heterogeneous characteristics and decision-making behaviors, thereby undermining analysis performance.

Agent-based model (ABM) has strong advantages in modeling decision-making behaviors (Liang et al., 2022). To this end, we built an agent-based model (ABM) with micro firm-level data to disentangle the uneven ETS policy effects on different types of firms and note their behaviors from a heterogeneity perspective. The following is a brief account of our model. There are two types of agents in an ETS with heterogeneous attributes: the government and firms. Firm agents pursue the maximum profits in the market through a series of decisions. Firms obtain permits from the government in every period and then make an annual production plan to maximize their profits according to the benefits from factors such as the commodity market, abatement cost, and carbon price. At the end of the period, firms will use the carbon reduction results to make decisions regarding the ETS, such as storing extra permits, trading in the ETS market, or accepting penalties. Firms upgrade their plans and decisions in the subsequent period according to the last period's results.

Our analysis generates three sets of managerial and policy insights:

- 1) Insights into the ETS market. Our core result is that the carbon price experiences a non-monotonic "L-shaped" trend, which initially remains at a low level for a long time and increases sharply after passing a threshold.
- 2) Insights into the ETS firms. Initially, most firms prefer to store extra permits rather than trade in the ETS. As the price increases, more firms participate in the ETS more actively.
- 3) Insights into ETS policy. As the carbon price increases, defaulting firms and market uncertainty also increase. ETS policies should create new strategies to cope with the risk of missing reduction targets.

Our contributions to the academic literature and managerial practice are threefold. First, we investigated ETSs using a bottom-up approach by focusing on firm-level decisions. Understanding firms' decision-making behaviors in an ETS can help uncover the underlying logic of the ETS and make policies more effective. Second, the diversity of firms was considered in the proposed model, providing insight into how different firm characteristics can influence ETS decisions. This can help us understand the ETS's emergence process and performance more accurately, including carbon prices, carbon reduction targets, and uncertainty. Furthermore, based on the simulation results, policymaking can be optimized toward greater accuracy and effectiveness. Third, unlike most previous studies based on virtual firm data, this study applies real firm-level data to the model. The characteristics of each firm are built from real firm data, thereby making the model solid and providing convincing results. The results show that the prediction of the ETS performance at an earlier time period is consistent with reality, which also verifies the advantage of the proposed model with real firm data. Based on these advantages, our model can be used as a policy simulation platform for mechanism design and can impact ETS evaluations.

The remainder of this paper is organized as follows: Section 2 presents an extensive literature review; Section 3 elaborates on the calibration and specifications of the simulation; the empirical results are presented and discussed in Section 4; and, finally, Section 5 concludes the study.

2. Literature review

This work overlaps with two research areas: research on the impact

of ETS on firm decision-making and behaviors and models that describe the processes of emissions permit prices.

2.1. The impact of ETS on firm behavior

Regulated firms in the ETS have dual roles as emitters and the trading body of emission quotas. A substantial number of studies have explored the impacts of ETS policies from the ETS firm's perspective. This strand of literature currently focuses on the following aspects: Impacts of ETS on the firm operation (Gong and Zhou, 2013; Krass et al., 2013; Li et al., 2018; Wang et al., 2018;), allowance trading decisions on carbon emission quotas (Chen et al., 2016; Zhang and Xu, 2013; Zhao et al., 2018;), and energy-saving technology investment (Fang and Ma, 2019; Tang et al., 2015; Zeng, 2019; Zhang et al., 2017). ETS would affect these aspects due to their emergence properties, and they could be the decisive factors influencing the carbon price trend.

Regarding the responses and business decisions of firms in the ETS, studies have explored the impact of the ETS on firms' optimal production quantity, energy consumption, optimal allowance price, and other economic indicators. For example, Purohit et al. (2016) studied the impact of the ETS quota and the allowance mechanism on firm inventory using a mixed-integer linear programming model in the face of unstable random consumer demand. Wang et al. (2018) assessed the impact of the cap-and-trade mechanism, carbon emission intensity, transaction cost, and other critical factors on firms' optimal production plans under the ETS. They noted that firms' optimal production and maximum profit were directly proportional to the carbon quota, but inversely proportional to carbon emission intensity and low-carbon processing costs. Chan et al. (2013) found that the EU ETS promoted the competitiveness of European power companies, which was mainly reflected in material costs and revenue.

As the key variable or parameter of the ETS mechanism, allowance trading strategies among firms have also attracted increasing attention from researchers. For example, Zhao et al. (2018) added fuzzy random variables as uncertain factors to estimate carbon prices. They explored how building material suppliers adjust their production and emission allowance purchase plans under different emission reduction policies. Zhang and Xu (2013) proposed a profit maximization model to characterize the optimal strategies of firms under an ETS. Specifically, they investigated the impacts of allowance caps and prices on factors such as carbon emissions and allowance trading. To assist firms in optimizing allowance transaction decisions, Chen et al. (2016) examined the behavior change in management decisions under the ETS. They explored the role of technological investment in reducing carbon emissions.

As a core policy objective of the ETS, technological innovation has been widely recognized as a critical factor in promoting the reduction of GHG emissions (Drake et al., 2016; Okushima and Tamura, 2010; Timilsina and Shrestha, 2009; Wei and Yang, 2010). For example, based on an uncertain technological learning optimization model, Fang and Ma (2019) found that the ETS induces the adoption and research and development (R&D) of emission abatement technology. Zeng and Zhu et al. (2019) studied the impact of an imperfect competition ETS model on the technological adoption of energy-intensive firms. The estimations showed that market power in such an ETS institution would lead to firms being prone to delay the adoption of new emission abatement technologies. When the demand for output is high, the diffusion of emission reduction technologies occurs early. Zhu et al. (2018) noted that regulated firms tend to overinvest in emission abatement technology, which may further lead to an oversupply of allowance in the ETS.

Based on the analysis mentioned above, the existing literature has explored the impacts of ETS on firms' decision-making behavior from various perspectives and has achieved fruitful findings. However, most of the literature is based on the theoretical underpinnings of the rational equilibrium framework, which strongly assumes perfect competition, rational expectation, and complete information. Consequently, these strong research assumptions lead to several limitations.

First, the rational equilibrium hypothesis ignores the dynamic interaction among ETS agents, especially considering the bounded rationality of firms (Yu et al., 2020). These limitations weaken the emergence properties and uncertainties of complex systems.

Second, few studies have focused on heterogeneity among ETS-regulated firms. Substantial differences exist among ETS firms, such as business scale and scope, allowance demand, emission abatement cost, technology preference, and market expectation (Brohé and Burniaux, 2015; Martin et al., 2011) have asymmetric effects. Therefore, it is natural to question whether the ETS has an uneven impact on different firms. Another underestimated issue is exploring the impact of the differences of such firms on ETS market equilibrium and market efficiency.

2.2. The use of agent-based models in ETS studies

The central strand of literature on emissions cap systems employs top-down mathematical models. Tang et al. (2020) reviewed the quantitative models used in ETS research, detailing how different models serve each ETS-involved agent and facilitate its ETS-related challenges. However, these top-down methods have certain defects. First, owing to the strong assumption underlined, such oversimplification overlooks the heterogeneity of industries and firms (Huang and Ma, 2016). Additionally, these models are often based on the unrealistic assumption that each entity has all the relevant information about peers' features and behaviors, which deviates from real behavioral dynamics under policy enactment (Rothkopf, 1999). Further, these methods neglect and fail to explore the dynamic interaction progress among stakeholders in simulating the equilibrium solution (i.e., optimality analysis) (Richstein et al., 2014). In an ETS, dealers jointly form the trading market in dynamic and complicated decision-making and interactive process (Weidlich and Veit, 2008).

Compared with the traditional top-down mathematical model, the ABM provides an explicit function of the system players' behavior. The "emergence properties" in a complex system can be explored from a bottom-up perspective (Tessfatsion, 2006). Because the ABM is equally effective at treating issues of interactive effect, heterogeneity, bounded rationality, and learning evolution, it is widely recognized as an effective tool for complex system modeling and analysis (Yu et al., 2020).

In the existing literature, the ABM has gained popularity in its projection of firms' long-term low-carbon transformation and evaluation of the ETS's policy effect. Chappin and Dijkema (2007) discussed the behavior decision of Dutch power firms under the EU ETS based on the ABM. They reported that firms must improve their power generation capacity by investing in new technologies to achieve the ETS emission reduction goal. To simulate the firms' production behavior in response to ETS, Yu et al. (2020) considered optional low-carbon technologies' production details and complexity in a multi-level time frame using the ABM. The simulation results indicated that a high emission reduction goal would lower the uncertainty of the subsidy price but would have a significant impact on the firm's output. Huang and Ma (2016) simulated the dynamic behavior of controlled firms under different quota rules. They found that increasing the air pollution cost in the carbon market limits the firms' trading decisions. Some studies on the Chinese ETS have recently attempted to combine the ABM with Monte Carlo simulations (Fang and Ma, 2019; Cong and Wei, 2010; Tang et al., 2015). Chen et al. (2018) used the combined model to capture the risk preferences and adaptive technology preferences in the low-carbon innovation process.

Zhu et al. (2018) explained the decision-making behavior of firms under three non-equilibrium frameworks by simulating regulated firms as heterogeneous agents with different allowance requirements, abatement costs, and technology preferences. A large body of literature has explored the impact of ETS on the behaviors of firms in various industries using the ABM. These industries include iron and steel (Li and Zhu, 2014; Zhang et al., 2011), petrochemicals (Li et al., 2020), electric power (Ringler et al., 2016; Zeng et al., 2020), and agriculture (Bakam and Matthews, 2009). ABMs view these energy-intensive industries as

dynamic interactions among agents and allow for insights into ETS operational rules.

3. Model specifications

3.1. Model framework

In this section, a novel ABM for the ETS is built to analyze the effect of the ETS and predict the carbon price using firm-level data. In contrast to the macro perspective of most previous studies (e.g., allowance policy, market mechanism), this proposed model investigated ETS policy and the decision-making behaviors of ETS firms from a micro perspective. Firms play an important role in the ETS. Both the carbon price and carbon reduction (the most important ETS index) are determined by firms. On the one hand, firms make decisions on their own carbon reduction, which can influence the total tradable carbon in the ETS. On the other hand, firms' trading strategies (i.e., buying, postponing/reserving, and selling allowances) can influence the carbon price of an ETS. Therefore, to deeply understand the mechanisms of ETSs, it is necessary to unveil the underlying logic at the micro level, especially with regards to the decision-making behavior of firms.

The agent-based model proposed in this study aims to analyze the ETS from the bottom-up by considering the decision-making behavior of firms. We can draw the mechanisms and operations of the ETS, including carbon trade, carbon price, and carbon reduction, and use this information to propose suggestions for ETS policy. The relationships between the micro- and macro-levels of the model are presented in Fig. 1.

The framework of the proposed model is illustrated in Fig. 2. There are two types of agents in the model: the government and firms. Under the ETS mechanism, each agent maximizes its profit as a rational economic man. They make various decisions on carbon reduction according to their aims, and interact with each other through the commodity and ETS markets. Each agent learns from the environment and historical data, thereby adjusting their decisions according to political and market changes. The government agency has the goal of carbon reduction in the abatement phase. It conducts three actions to achieve the goal: allocating allowance to firms, supervising the ETS market, and penalizing non-compliant firms. The firm agents pursue maximum profit in the market through decisions such as investing in carbon reduction technology, updating production plans, storing extra carbon, trading in the ETS market, and accepting penalties.

It is worth noting that there are two systems in the model. One is the economic system (shown in yellow) related to production and monetary benefit/cost. The other is the carbon system (shown in green) related to carbon emissions and carbon trade. These two systems are integrated into a holistic approach in the ETS market. Furthermore, the capital flow

(shown by the solid black line) and carbon flow (indicated by the black dotted line) form a large circulation path within the national economy.

The proposed ABM can be divided into several modules, as shown in Table 1. The entire framework and each module will be discussed in subsequent sections.

3.2. Agent definitions

3.2.1. Government agent

The government plays an essential role in the ETS by implementing and supervising the policy to achieve the carbon reduction target (Tang et al., 2020) (e.g., the promise in the Paris Agreement). To achieve this, the government agent conducts three main actions (shown by the red line in Fig. 2):

- 1) Breaking the long-term target into a yearly goal and transforming it into a carbon allowance for each firm in different industries.
- 2) Building up the ETS market, creating the ETS market rules, and supervising the operation of the ETS market (no direct price control).
- 3) Establishing the punishment mechanism and penalizing firms that do not meet the standard.

3.2.2. Firm agent

Firms are the main participants in both traditional commodity and ETS markets. As economic agents, each firm aims to maximize its profit (Zhu et al., 2018). In the conventional commodity market, a firm must only consider the costs and benefits of production. After the ETS policy, a firm must balance more factors, including carbon allowance, investment in carbon reduction, and current and future carbon prices. To achieve this, firm agents must make the following decisions:

- 1) Should the firm invest in carbon reduction or just buy allowances from the ETS market?
- 2) Should the firm buy allowances in the ETS or accept the penalty directly?
- 3) Considering the marginal abatement cost (MAC), which is more beneficial, buying or selling carbon allowances in the ETS?
- 4) What is the trend in carbon prices? According to this trend, will selling extra carbon permits be beneficial in the current period or in the future?

3.2.3. Agent attributes

To model the government and firms of the real world, their attributes were projected onto the agent's variables in the model, as shown in Table 2. The first two variable types are the attributes of government and firm agents. Along with the variables for each agent, there were environmental and intermediate variables. Environmental variables describe external environmental conditions (e.g., carbon price). Intermediate variables were applied to facilitate the operation of the model.

3.2.3.1. Firm attributes. Industry category Cat : This variable represents the industry category of the firms. The industry category determines the marginal abatement cost (MAC) and the emission intensity ($carbon_int$) of a firm because emission reduction technologies vary among industries. The industry category in this study was obtained from the Chinese industrial enterprise database in 2013.

Marginal abatement cost MAC : This variable represents the marginal abatement cost of firms. Each firm has a particular MAC curve, mainly influenced by industry category Cat . The MAC curves in this study are from the MACs in different industries obtained from the literature (Xiao et al., 2017).

Emission intensity $carbon_int$: This variable represents a firm's emission intensity in the initial stage. Each firm has a particular emission intensity, which is mainly influenced by the industry category Cat . The $carbon_int$ used in this study was obtained from the literature (Yu

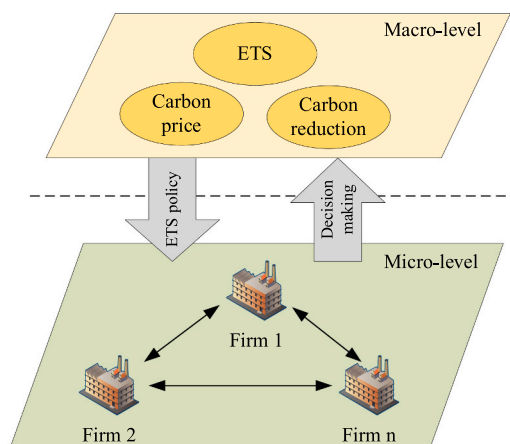


Fig. 1. The schematic of the bottom-up model for ETS.

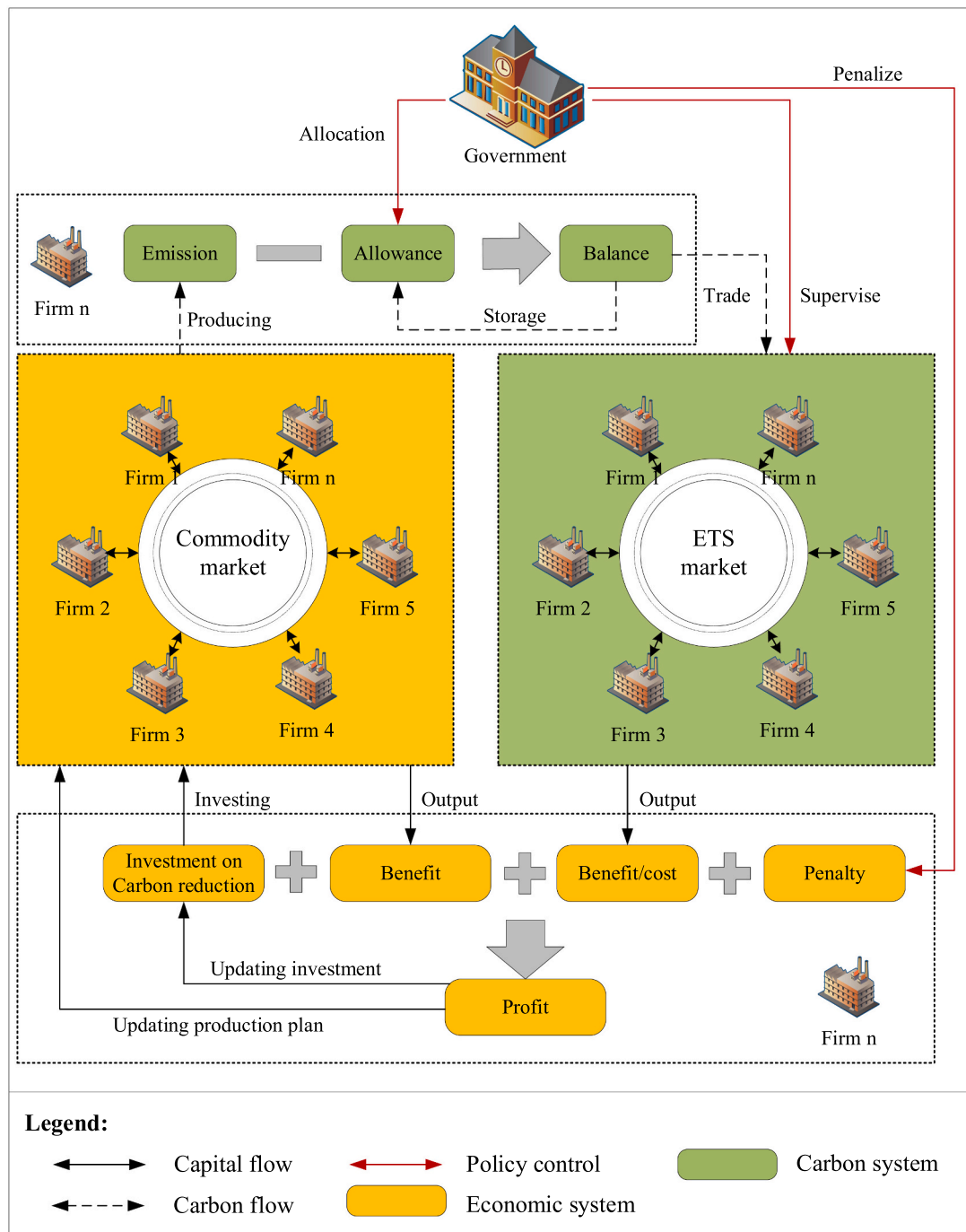


Fig. 2. The framework of the agent-based model for ETS.

et al., 2018).

Output value/output: This variable represents a firm's yearly output value. In this study, this attribute is initially from the Chinese industrial enterprise database for 2013; after that, it is influenced by a firm's decision to reduce emissions.

Carbon emissions/carbon_emi: This variable represents a firm's yearly carbon emissions. In the initial stage, this attribute is the product of output value (*output*) and emission intensity (*carbon_int*). Thereafter, this attribute will be further influenced by a firm's decision to reduce emissions (*Tec_D*).

Carbon balance/bal: This variable represents the yearly carbon balance of a firm, which is equal to carbon emissions (*carbon_emi*) minus allowances (*A*).

Profit/pro: This variable represents a firm's annual profit. This attribute is determined by profits from the commodity and ETS markets. The former is based on the firm database, and the latter is based on firms' actions (i.e., to buy, sell, store, and receive penalty) and the carbon price in the ETS. Because a firm aims to maximize its profit, profit will also influence the firm's decisions in the following year.

Decision on carbon reduction/Tec_D: This variable represents the decisions on whether to invest in carbon reduction technology, which is mainly influenced by the marginal abatement cost (MAC) and carbon price (*p*). The decision-making process will be discussed in Section 3.2.5.

Decision on actions in ETS/Trade_D: This variable represents the decisions on an action in the ETS market, including to buy, sell, store, and receive penalties. The decision is influenced by the penalty (*Pen*),

Table 1
Design of the agent-based model at a module level.

Perspective	Introduction
(A) Agent definition	
Government	Government allocates allowance to firms and supervise the market
Firm	Firms with heterogenic attributes pursue maximum profits in the market
(B) Agent interaction	
Allowance	Government allocates allowance to firms
Trading in ETS	Trade among firms in ETS based on the clearing price mechanism
Punishment	Government penalizes non-qualified firms
(C) Agent behavior	
Emission reduction	Decide whether to reduce emissions based on the historical outputs
Trading in ETS	Decide out of four choices (sell, buy, storage and penalty)
(D) Rationality	
	Agents follow the rational behavior assumption, making decisions to maximize their profits
(E) Learning	
	Firms will learn from the historical data of firm operation and the ETS market

Table 2
Variable definitions in the model.

Type	Variable	Definition
Firm	<i>output</i>	Yearly output value of a firm
	<i>pro</i>	Yearly profit of a firm
	<i>carbon_int</i>	Emission intensity of a firm at the initial stage
	<i>carbon_emi</i>	Yearly carbon emission of a firm
	<i>MAC</i>	Marginal abatement cost of each industry
	<i>bal</i>	Yearly carbon balance of a firm
	<i>Tec_D</i>	Decision on whether to invest in carbon reduction technology
	<i>Trade_D</i>	Decision on an action in ETS market
	<i>Cat</i>	Industry category of a firm
	<i>A</i>	Allowance to each firm
Government	<i>RR</i>	Target rate of carbon reduction
	<i>Pen</i>	Penalty for non-qualified firms
ETS	<i>p</i>	Carbon price in ETS

carbon balance (*bal*), carbon price (*p*), and profit (*pro*). The decision-making process will be discussed in Section 3.2.5.

3.2.3.2. Government attributes. Rate of carbon reduction*RR*: This variable represents the target rate of carbon reduction. This attribute is determined by the overall carbon reduction target and is stable during a particular period. The value of this attribute is derived from the real target of the Chinese government (details in Section 3.4).

Allowance*A*: This variable represents the allowance of each firm. The allowance for each firm is based on the government's carbon reduction target (*RR*) and the historical carbon emissions of the firm (*carbon_emi*).

Penalty*Pen*: This variable represents the government penalties. The penalty is an essential aspect of ETS policy. The value of this attribute is derived from the real penalty policy of the Chinese government (details in Section 3.4).

3.2.3.3. ETS attributes. Carbon price*p*: This variable represents the ETS carbon price. The carbon price is based on firms' carbon emissions (*carbon_emi*) and ETS actions (*Trade_D*). The pricing mechanism will be discussed in Section 3.3.

3.2.4. Agent interaction

Fig. 3 shows the agent interactions and variable connections of the proposed ABM. The large boxes represent the two agent types (i.e., firm and government) and the ETS market. Agents interact with each other

through their actions and changes in the variables. The variables of an agent can influence other variables and can also be influenced by other variables. For example, the firms' decisions on actions in the ETS market (*Trade_D*) will influence carbon prices (*p*). Conversely, the carbon price (*p*) will affect the profits of firms (*pro*) and further influence their decisions on actions in the ETS market (*Trade_D*) in the next stage.

There are three categories of variables (i.e., action, static, and action & time-dependent) shown by different shapes (Fig. 3). Static attributes are parameters that do not change over time. For example, the carbon reduction target (*RR*) is constant during a given period. The action attributes represent the decision-making behavior of the agents. In this study, the two main actions are decisions on carbon reduction (*Tec_D*) and ETS actions (*Trade_D*). The action- and time-dependent attributes change based on time and the agents' actions. For example, a firm's carbon emissions depend on its yearly output value (*output*), carbon intensity (*carbon_int*), and decisions on carbon reduction (*Tec_D*).

3.2.5. Decision-making behavior

3.2.5.1. Decision-making process. Fig. 4 shows the agents' decision-making process. The process is shown in pink, illustrating the entire workflow of an agent. At the beginning of a year, a firm will receive an allowance from the government, and then the firm will draw up an annual production plan. In this plan, a firm needs to decide whether to reduce carbon emissions through technical progress. Subsequently, the firm will implement the plan throughout the year. The total carbon emissions are audited, and the carbon balance is calculated at the year's end. If the real carbon emission exceeds the allowance, a firm will go to Decision 3 to decide whether to buy an allowance from the ETS market or accept the penalty. If the real carbon emissions are less than the allowance, a firm will go to Decision 4 to decide whether to sell the extra carbon allowance in the ETS market or store the extra carbon allowance for a future stage. A firm will then pay a cost or benefit based on this decision. Finally, a firm will calculate the total profit for this year and start another loop in the next year; this year's resulting profit will influence the next year's decisions.

Along with the process part, there are function and data parts, in green and orange, respectively (see Fig. 4). The function part divides particular functions into modules and supports the workflow; for example, the functions to model the MAC curve, calculate the profit, and optimize a firm's strategy. The data part stores various data and supports the process and function parts. The data mainly includes the firm's emission, technical progress, policy, and market data.

3.2.5.2. Decision on carbon reduction. At time *t*, the output value of firm *j* in sector *i* is defined as $output_{i,j,t}$. It depends on a firm's current business performance, regarding the profit changes in the last stage, by following the learning principle (Liu et al., 2012).

$$output_{i,j,t} = \begin{cases} output_{i,j,t-1} + \Delta output_t, & \text{if } pro_{i,j,t-1} - pro_{i,j,t-2} > \delta \\ output_{i,j,t-1}, & \text{if } |pro_{i,j,t-1} - pro_{i,j,t-2}| \leq \delta \\ output_{i,j,t-1} - \Delta output_t, & \text{if } pro_{i,j,t-1} - pro_{i,j,t-2} < -\delta \end{cases} \quad (1)$$

where $pro_{i,j,t}$ represents the profit of firm *j* in sector *i* at time *t*, and the non-negative parameters $\Delta output_t$ and δ denote the output variation and a given threshold, respectively. Since China is in a stable economic environment, the variation is very limited in the case study. A firm increases its output when its profit grows beyond the threshold in the last period. In contrast, a firm decreases its output when the profit declines by more than the last period's threshold. A firm maintains the same output when the profit does not change significantly. The value of δ refer to previous literatures for Chinese ETS (Tang et al., 2017; Yuan et al., 2018; Zhu et al., 2019), which is empirical value validated by previous studies.

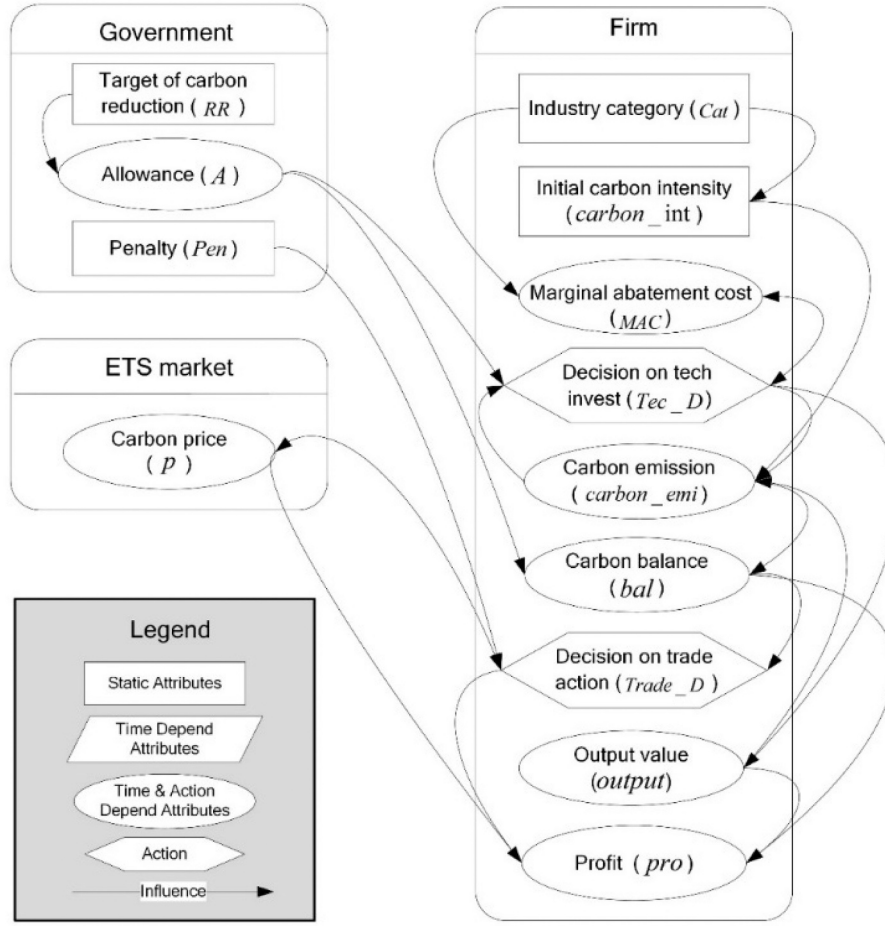


Fig. 3. Variables and interactions between the government and firms.

The carbon emissions of firm j in sector i at time t are defined as $carbon_emi_{i,j,t}$, which is determined by $output_{i,j,t}$ and carbon intensity $carbon_int_{i,j,t}$.

$$carbon_emi_{i,j,t} = output_{i,j,t} * carbon_int_{i,j,t} \quad (2)$$

At the beginning of the year, a firm can predict the year's carbon emissions. Based on the predicted carbon emissions $carbon_emi_{i,j,t}$, the firm can decide whether to reduce carbon emissions this year according to Eq. (3). If the predicted carbon emissions are more than the allowance of firm j , and the MAC of firm j is less than the ETS' carbon price, the firm will reduce its carbon emissions (Decision 1 in Fig. 4). Otherwise, the firm will not reduce its carbon emissions (Decision 2 in Fig. 4). For example, if $MAC_{j,t} > p_t$, it is more beneficial to buy a carbon allowance in the ETS than to reduce carbon emissions.

$$\begin{cases} \text{Decision 1,} & \text{if } carbon_emi_{i,j,t}^p > A_{j,t} \text{ and } MAC_{j,t} < p_t \\ \text{Decision 2,} & \text{otherwise} \end{cases} \quad (3)$$

3.2.5.3. Decision on action in ETS. After making a decision, a firm goes to the next stage, implementing the plan and producing the real total carbon emission $carbon_emi_{i,j,t}$ according to Eq. (4). The carbon balance of firm j in the sector i at time t $bal_{i,j,t}$ can be calculated using Eq. (4).

$$bal_{i,j,t} = carbon_emi_{i,j,t} - A_{j,t} \quad (4)$$

Based on $bal_{i,j,t}$, firm j can choose between Decisions 3 and 4 according to Eq. (5).

$$\begin{cases} \text{Decision 3,} & \text{if } bal_{i,j,t} > 0 \\ \text{Decision 4,} & \text{otherwise} \end{cases} \quad (5)$$

If $bal_{i,j,t} > 0$, firm j will decide whether to buy carbon allowances from the ETS market or to accept the government's penalty. This depends on which cost is higher, as shown in Eq. (6). If the penalty is greater than the cost of buying in ETS $bal_{i,j,t} * p_t$, firm j will buy carbon allowances in the ETS. Otherwise, firm j will not purchase from the ETS and will accept the government penalties.

$$\begin{cases} \text{Decision 3,} & \text{if } pen > bal_{i,j,t} * p_t \text{ and successful trade in ETS} \\ \text{Decision 4,} & \text{otherwise} \end{cases} \quad (6)$$

If $bal_{i,j,t} < 0$, firm j will decide whether to sell carbon allowances in the ETS market or store the extra the carbon allowance for the next stage. This depends on the allowance reduction, and whether the benefit is sufficiently high, as shown in Eq. (7). If the extra allowance is less than the allowance reduction in the next stage and the benefit of selling allowance is high compared to the total profit, firm j will sell the carbon allowance in the ETS. Otherwise, firm j will store the carbon allowance for the next stage.

$$\begin{cases} \text{Decision 5,} & \text{if } |bal_{i,j,t}| > A_{j,t} - A_{j,t+1} \text{ and } |bal_{i,j,t}| * p_t > pro * \theta \\ & \text{and successful trade in ETS} \\ \text{Decision 6,} & \text{otherwise} \end{cases} \quad (7)$$

where $A_{j,t+1}$ is the allowance in the following year and θ is the threshold of the benefit proportion. After all, firms make their decisions and trade in the ETS market, and they can calculate their total profits in time t . Thereafter, the model will enter the next loop, time $t + 1$.

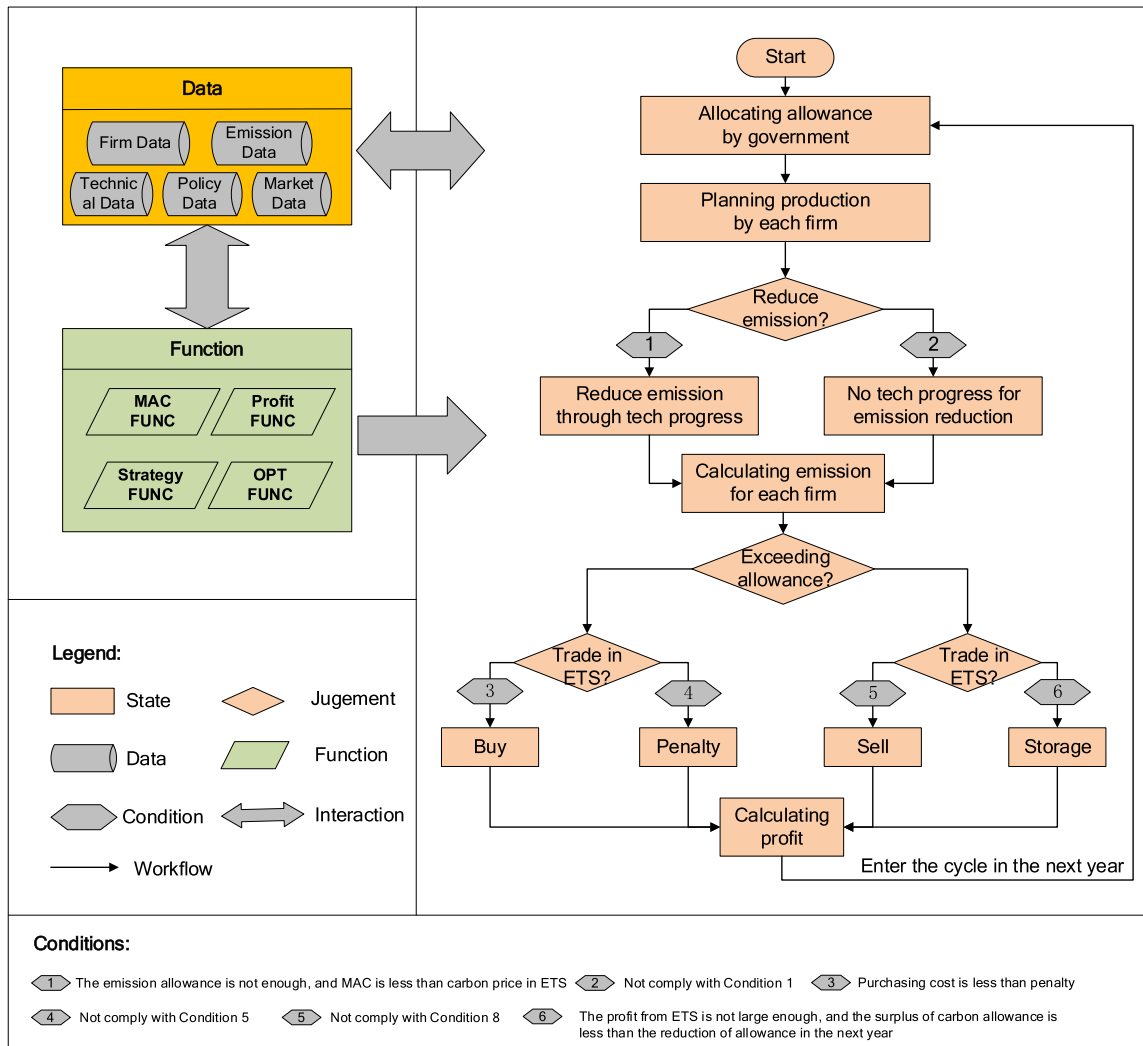


Fig. 4. The agents' decision-making process in the emissions trading scheme (ETS).

3.3. Market mechanism

The equilibrium carbon price is the clearing pricing in the ETS (Tang et al., 2017; Yuan et al., 2018). Carbon allowances in an ETS can be thought of as goods in the commodity market. According to microeconomics, the clearing price is determined by supply and demand in the market; it assumes that, in a market, the unit price for a particular good will vary until it settles at the point where the quantity demanded (at the current price) equals the quantity supplied (at the current price), namely, $Q^{buy}(p) = Q^{sell}(p)$. The demand quantity ($Q^{buy}(p)$) and supply quantity ($Q^{sell}(p)$) depend on firms' decisions (i. e., Decision 1 - 6 in Section 3.2.5).

Moreover, there are two constraints on carbon price. One constraint is for the seller: $p_{j,t}^{sell} > MAC_{j,t}$. This means that the seller's offered price $p_{j,t}^{sell}$ must be higher than the firm's $MAC_{j,t}$, otherwise, the seller will suffer a loss in the trade. The other constraint is for the buyer, namely, $p_{j,t}^{buy} * carbon_emij_t < Pen$. This means that the purchase cost $p_{j,t}^{buy} * carbon_emij_t$ must be lower than the government's penalty; otherwise, the buyer cannot benefit from the trade.

Because of the complexity of $Q^{buy}(p)$ and $Q^{sell}(p)$, the equilibrium carbon price (clearing price) of the ETS cannot be expressed by closed-form functions, and the analytical solution cannot be derived. However, with the assistance of simulations, we can obtain numerical solutions based on the equilibrium condition. In each time step (t), the agent-based model can simulate the firms' decision-making in the ETS

and generate the supply or demand quantity for each firm in the ETS, $Q_{i,t}^{buy}(p)$ and $Q_{i,t}^{sell}(p)$. Based on these results, the simulation platform can draw supply and demand curves and identify the equilibrium price and quantity in the ETS under these constraints.

3.4. Model calibration and simulation setting

This study applies the proposed ABM to the ETS in Shanghai, China. The pilot ETS in Shanghai was launched in 2013, and 197 firms were included initially, mainly from the iron and steel, chemical, cement, energy, light, and other industries (e.g., building and service) (Wu et al., 2014). Because carbon emissions from other industries are very low, this study only includes 128 firms in the top five industries. The firm-level data of the 128 firms are from the Chinese industry enterprise database for 2013, and includes each company's industry, output value, and profit.

This study used real parameters based on the ETS policy for Shanghai's pilot ETS. The carbon reduction target rate is approximately 2% per year, and the penalty for each unqualified firm is 100 thousand CNY per year. According to Shanghai's ETS policy, the allowance for each firm is based on the historical carbon emissions minus the carbon reduction target in each year. The carbon intensity and MAC of each industry refer to the references for industrial carbon emissions. Firms' carbon intensity and MAC vary based on the industrial data following a normal distribution $N(0, 0.01)$.

This study applied 10 simulated years, from 2017 to 2026. Because low-carbon technology generally has a long investment cycle, our study selected a 10-year period to simulate the dynamics of carbon prices. Following the practices of Abrell et al. (2021), who investigated the allowance-trading behavior of the ETS in Europe, a 10-year period indicates that the ETS has moved from the establishment stage to the maturity stage. The simulation improves its precision through one hundred repetitions, averaging the results. Table 3 presents the simulation settings.

4. Empirical results

4.1. ETS performance

4.1.1. Equilibrium carbon price

From the perspective of time, the carbon price experiences an L-shaped trend, which remains low for an extended time but rises rapidly after an inflection point (around the 8th year), as shown in Fig. 5. The carbon price remained low in the first seven years, at approximately 50 CNY. After that, the carbon price increased sharply. It was around 150 CNY in the 8th year, 250 CNY in the 9th year, and 700 CNY in the 10th year.

4.1.2. Carbon reduction

Fig. 6 shows the rate of total carbon reduction in the Shanghai ETS. The carbon reduction rate can be increased by government setting (2% per year), but the uncertainty increases in the last periods (9th and 10th years). This indicates that the risk of missing carbon reduction targets will increase remarkably by the end of ten years.

Fig. 7 shows the relationship between the carbon price and the rate of carbon reduction. The results show that the uncertainty of carbon reduction also increases with an increase in the carbon price. Thus, when the carbon price fluctuates significantly in the late stage, it becomes challenging to ensure the achievement of the carbon reduction target.

4.2. Firm performance in ETS

4.2.1. Firms' benefit from ETS

Fig. 8 shows the benefit that firms receive from ETSS. The steel and chemical industries started to have higher profits in the ETS market, but the fluctuations were relatively large in the later period. This is mainly due to the difficulties associated with falling MACs in the late-stage, which led to greater cost pressure on carbon reduction. The other three industries (i.e., cement, power, and light industries) have maintained a relatively stable amount of benefit from ETSS, because their MAS curves are relatively flat.

4.2.2. Firms' carbon reduction

Firms in the chemical industry have the most significant carbon emission reductions among industries. This is followed by steel in the middle, whereas the amounts for the cement, power, and light industries are relatively small. From a time perspective, the carbon reduction of firms in the chemical industry decreased significantly in the late stage. In contrast, the other industries maintained stable carbon reductions

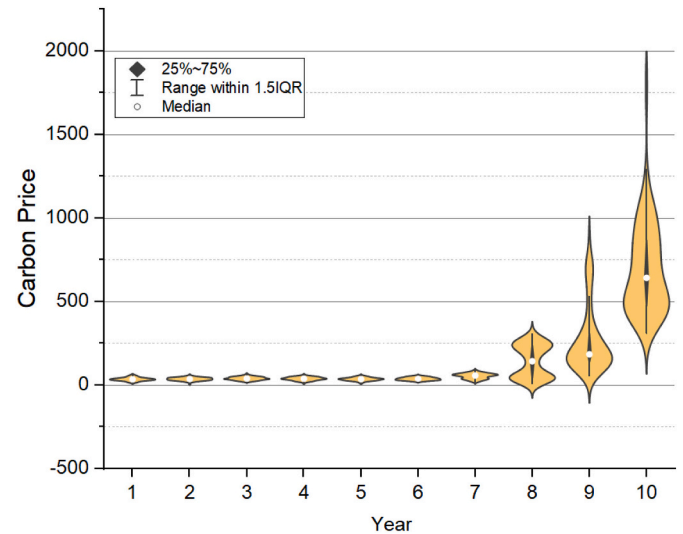


Fig. 5. Carbon price in the emissions trading scheme (ETS).

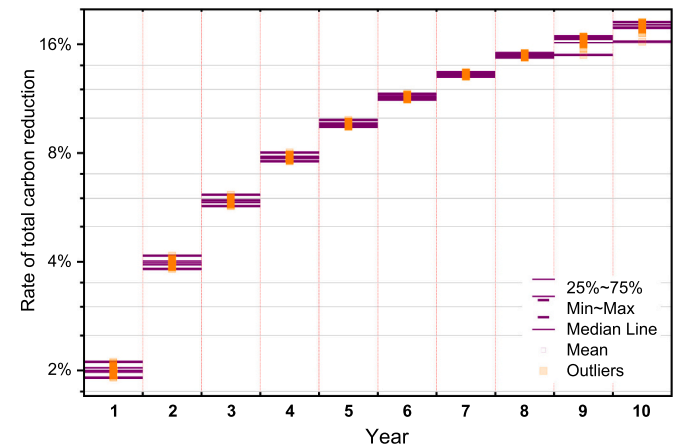


Fig. 6. Rate of total carbon reduction.

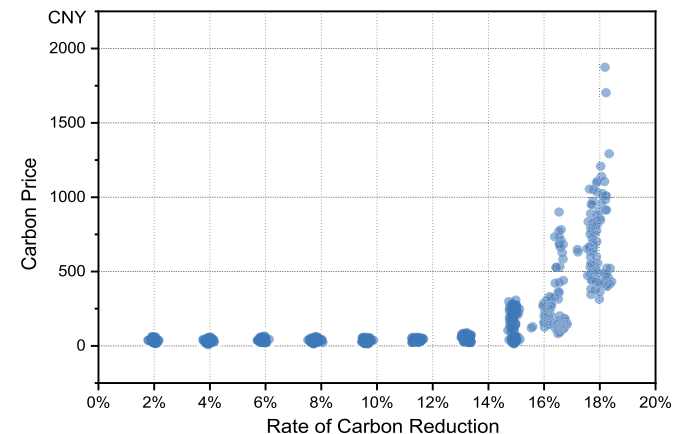


Fig. 7. Relationship between carbon price and carbon reduction rate.

Table 3

Simulation settings on the agent-based platform.

Variable	Symbol	Unit	Value
Firm number	N_j	NA	128
Industry number	N_i	NA	5
Penalty	Pen	CNY	100,000
Reduction rate	RR	NA	2%
Simulation time span	T	year	10
Repeat times	RT	NA	100

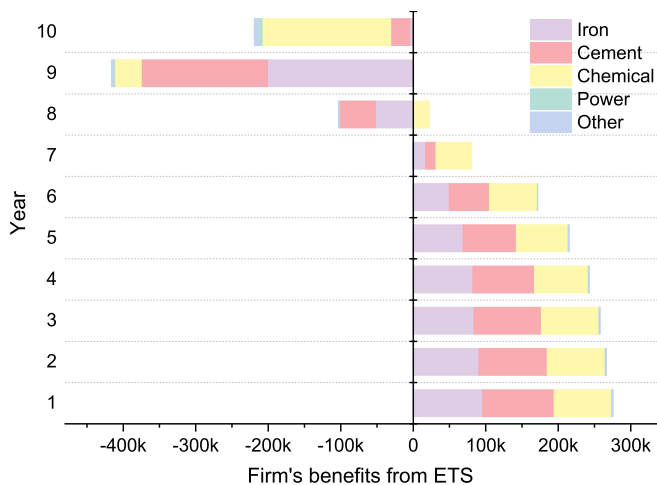


Fig. 8. Firms' benefit from the emissions trading scheme (ETS).

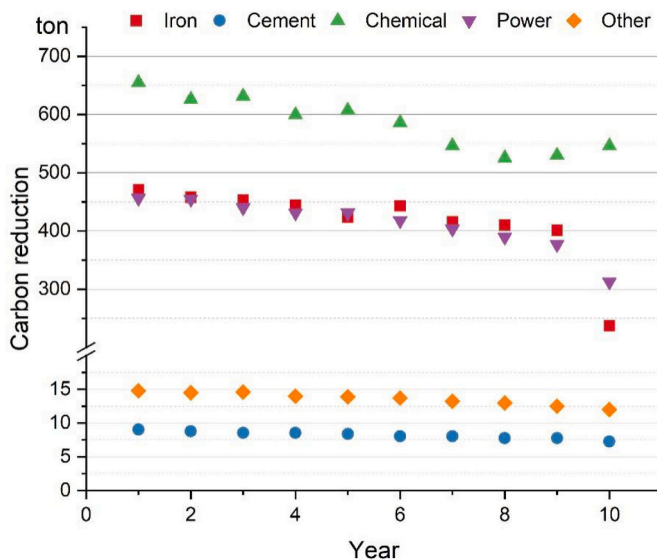


Fig. 9. Firms' carbon reductions.

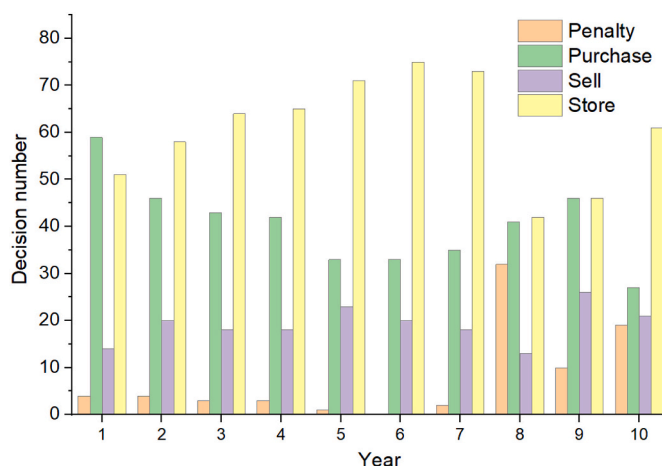


Fig. 10. Firms' decisions in the emissions trading scheme (ETS).

during the ten years, as shown in Fig. 9.

4.3. Firms' decisions in ETS

4.3.1. Firms' decisions

Fig. 10 shows the firms' overall decisions. The vertical axis represents the number of firms making a decision, and the horizontal axis represents the simulation year. As shown in Fig. 10, very few companies chose to receive penalties during the ten years. The number of firms choosing to sell was relatively stable and rose slowly at the end of the ten years. Although many firms have surplus carbon allowances in the initial stage, they will store the allowance because of low carbon prices and meager profits from the ETS market. At a later stage, the number of companies choosing storage will decline.

In contrast, the number of companies choosing to buy decreases at the beginning but increases at a later stage. The storage and purchase curves form a "scissor cross". This indicates that with the development of the ETS, the demand for carbon allowance will increase, and the supply of carbon allowance will decrease. This can also explain why carbon prices increase sharply at the end of the ten years.

4.3.2. Decisions in different industries

Fig. 11 (a) shows firms' purchase decisions in different industries. Among the companies that chose to purchase carbon allowances, the chemical industry was the leader initially, and light industries rapidly increased purchases by the end of the decade, exceeding those of the chemical industry. This is because the light industry has lower carbon emissions per unit GDP, and much higher MAC (Choi et al., 2012; He, 2015; Xiao et al., 2017; Zhou et al., 2015). The MAC difference between heavy and light industry is more than ten times, which brings more challenges for light industry to obtain a similar carbon decline compared to other industries.

Fig. 11 (b) shows firms' selling decisions in different industries. Firms in the chemical industry also led among those who chose to sell allowances, especially at the end. The number of firms in the chemical industry further increased in the end and contributed to a significant share of selling decisions.

Fig. 11 (d) shows firms' decisions to accept penalties in different industries. Not all industries chose to be punished in ETS' early stages from an industrial perspective. Still, the steel and power industries had a relatively higher proportion of those who chose to be punished in the later stages.

5. Discussion

5.1. Carbon price and carbon reduction in ETS

According to the results in Section 4.1, the carbon price remained at a shallow level for a long time at the beginning of the ETS, and then increases sharply after passing a threshold. The carbon prices of ETS in China remain in low level in recent years. Shown as Table 4, the mean carbon prices from 2014 to 2018 are around 35–38 CNY/ton. It is basically consistent with the results of this study, which also validate the rationality and dependability of the proposed model. It is worth noting that the carbon price of Shanghai ETS dropped significantly in 2015, because the market mechanism in Shanghai ETS was changed. In 2013, the local government announced different allowances are for three years (2013–2015), which are SHEA13, SHEA14, and SHEA15. In 2016, the market would transition into a single SHEA, which can be cross-year traded. Therefore, the SHEA15 was in relatively low price, but the carbon price recovered quickly after 2017.

This has also occurred in the developed world's ETS. As the first mandatory carbon trading market in the United States, the RGGI experienced a downturn in the initial carbon market. Since 2014, the carbon market has gradually ventured onto the right track, with continuously rising carbon prices. By June 2021, the average price had reached

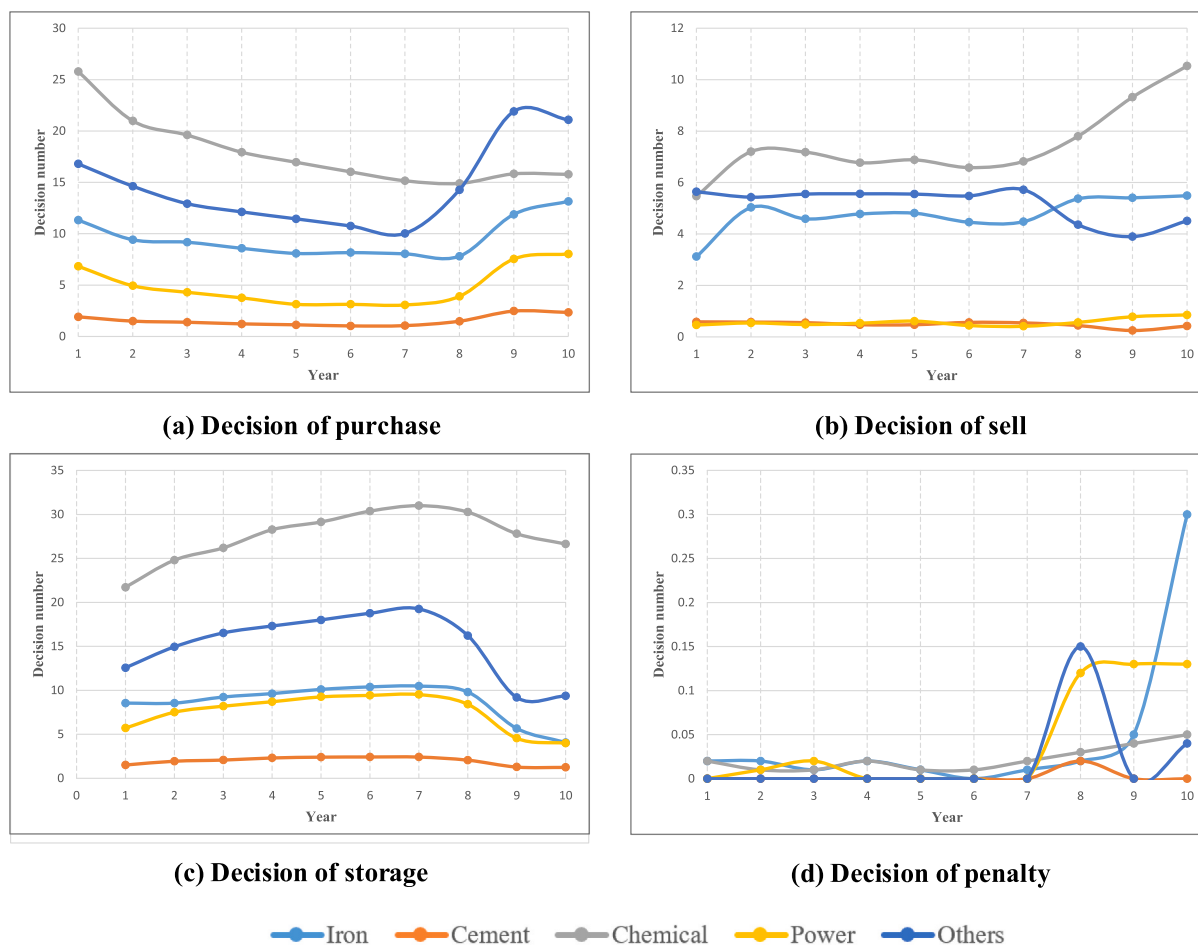


Fig. 11. Firms' decisions in different industries.

Table 4
validation by historical carbon price in Shanghai ETS.

Year	Historical carbon price in Shanghai ETS (CNY)		Mean carbon price in simulation (CNY)
	Min	Max	
2014	28.0	44.9	37.8
2015	9.5	35.9	36.6
2016	4.2	27.2	35.3
2017	24.8	39.4	36.7
2018*	27.8	42.6	37.0

* In 2018, the Chinese government announced that the national ETS would be prepared in Shanghai, which resulted in significant changes to the framework and mechanism of the ETS in Shanghai since then. Therefore, we included real observations of the ETS from 2014 to 2018, which represents a relatively stable period.

\$7.97/ton. This is because technical progress is relatively easy, and MAC is very low initially. Carbon reduction was a low-hanging fruit during this period; thus, the carbon price was low. After firms pick the low-hanging fruits, it is difficult to reduce carbon emissions. Firms must invest more in technical progress and equipment upgrades. Therefore, the carbon price tends to surge in subsequent periods.

A drastic increase in carbon price will negatively impact both firms and the carbon reduction targets of the ETS. For firms, the uncertainty of carbon prices leads to difficulties in decisions regarding technical progress and heavy carbon reduction loads, thereby impacting industrial and economic development. For ETS, drastic fluctuations in carbon prices may lead to high market risk. Furthermore, some firms may miss

reduction targets, impacting China's overall reduction target. Therefore, the government should be fully prepared for this risk and optimize the market mechanism to respond to fluctuations in carbon prices.

5.2. The diverse decisions of firms

Each firm has differentiated decisions according to different factors, including industries, production values, MAC curves, and carbon prices. For example, purchase and storage decisions presented double scissor movements during the ten years. Initially, purchase decisions are more than storage decisions. Because the carbon price remains low, more firms choose to store carbon allowances in the future. As the carbon price increases, more firms sell rather than store carbon allowances in subsequent periods. In contrast, previous studies are mostly based on equilibrium models, which usually simplify the heterogeneous behavior of firms into unified choices. The assumption of a representative firm or sector limits the model in distinguishing the differences within industries and periods and further weakens the ability to capture the emergence properties and uncertainty of the system.

According to the current policy, all firms in different industries are required to achieve the same carbon reduction target (with some minor adjustments) and use the "Grandfather's method". However, MAC curves are significantly diverse among different industries. If the government creates a unified policy for different industries, it will cause firms in different industries to have different carbon reduction loads. For example, it is easier for firms in the power industry to reduce their carbon emissions by 2% than firms in other industries. This hinders the development of some industries and leads to an imbalance among

industries. According to the low-carbon technology gap, the government should adopt suitable mitigation policies for various sectors and firms to avoid “one size fits all” policies.

5.3. Supervision and penalties

Supervision and penalties are important mechanisms in the ETS. According to the results in Section 4.3, few firms choose penalties in the early and middle stages, which is consistent with actual situations. According to relevant reports, compliance with China’s pilot carbon market is more than 97%, and Shanghai’s regulated firms have achieved 100% compliance for seven consecutive years. However, as the carbon price increases, more firms choose to receive the penalty rather than reduce carbon emissions or buy allowances from the ETS. This indicates that these punished firms missed their targets in that year.

According to the current policy, the penalty is only 100,000 CNY in the Shanghai ETS. On the one hand, the penalty is too limited for ETS firms, and the production values of most firms exceed 100 billion CNY. For these large-scale firms, 100,000 CNY is almost negligible from an economic perspective. Therefore, more firms take the penalty in the later stage and cause high uncertainty in achieving the reduction target.

On the other hand, the penalty lacks diversity as it is the same for all firms, but its industry attributes, production values, and MAC are significantly different. For example, firms’ production values range from 0.1 to 10,000 billion CNY. Thus, it is unreasonable to set a unified penalty for firms with considerable differences. The government should design penalties adapting to different types of firms rather than this “one-size-fits-all” policy. In contrast, the UK’s strict and diversified punishment mechanism provides useful insights. Relevant departments punish unrelated firms with economic, behavioral, reputational, and legal consequences. More importantly, the UK market makes a detailed distinction between each situation to ensure the feasibility and accuracy of its implementation.

6. Conclusions

This study proposes an empirical ABM to analyze firms’ decision-making behaviors in the ETS. This study predicted the disaggregated firms’ decisions and the Shanghai ETS’s emergence performance over ten years based on the proposed model. This study had several important findings. First, the carbon price experiences a non-monotonic “L-shaped” trend. During the early stage, the carbon price maintains an initial low level because technical progress is relatively easy and the MAC is low. After crossing a threshold, it becomes difficult for enterprises to reduce carbon emissions; thus, the carbon price increases sharply. Second, as the carbon price increases, more firms sell rather than store carbon allowances during the subsequent periods. Transactions in the ETS become active, particularly among low-emission firms. Third, the current ETS penalty lacks deterrence and diversity, which is inadequate for inducing technological progress and may cause high uncertainty in achieving the reduction target. Unlike previous models based on the equilibrium assumption, our study focuses on the heterogeneity of different industries and captures the emergence properties and uncertainty of the ETS.

From these research findings, this study collated the following policy implications. First, it is important to adopt innovative financial institutions and derivatives. Firms in light industry will face great challenges in carbon reduction due to their features of MAC. Compared to heavy industry, firms in light industry can have very limited profits from permit speculation. They treat trading in ETS more as a necessity for life than as a primary source of profit. This lukewarm approach will negatively impact ETS trading activity. Therefore, specialist agencies, especially large banks and hedge funds, which will be more active in trading permits, should be encouraged to join the ETS. Second, the research results show that enterprises in different industries have different decision-making behaviors in the carbon market, and will also obtain

varied emission reduction effects and benefits. It is because firms in different industries have heterogeneous attributes and MAC curves, that is, for the same proportion of emission reductions, firms have to pay significantly different MACs. However, ETS mainly use Grandfathering method, which is a “one-size-fits-all” policy. It only allocates permits to firms based on their historical emissions, but neglects differences among firms. To improve efficiency of ETS, it is important to formulate a new permit allocation standard and consider MACs of firms in different sectors, thereby providing a tailored permit allocation method. Third, protective policies are recommended to promote market development regarding ETS trends. In the later stage, high price volatility is not conducive to the stable development of the ETS. It is necessary to enact a market-stabilization mechanism to reduce the risk of price volatility, which is essential to alleviate the economic burden on firms caused by excessive carbon prices. Finally, ETS’s well-considered and tailored supervision and penalties are desired by accounting for the significant heterogeneity among regulated firms and industries. The government should design customized penalties adapting to different types of firms, concerning different sizes, sectors, and stages, rather than “one-size-fits-all” policies.

Concerning research limitations, the proposed model does not capture learning effects and interactions among firms. However, firms can learn from each other through previous benefits and market information. For example, a firm may choose to upgrade carbon reduction technology and sell allowances in an ETS after other firms have benefited from this approach. In future studies, the model should be enhanced to investigate the influence of the learning effect and the interactions among firms. Furthermore, the proposed model currently assumes that the macro environment remained stable for ten years, including the economic environment, technology development, and market mechanism of the ETS. This limits the scope of this study. In future studies, a sensitivity analysis can be implemented to test how environmental factors influence the results, facilitating the improvement of results and policy implications.

CRedit authorship contribution statement

Yigang Wei: Conceptualization, Writing – original draft, Funding acquisition. **Xin Liang:** Methodology, Writing – original draft, Funding acquisition. **Liang Xu:** Visualization, Writing – review & editing. **Gang Kou:** Validation, Writing – review & editing. **Julien Chevallier:** Resources, Writing – review & editing.

Declaration of Competing Interest

None.

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